BT 201 Final Current Solved – Muhammad Nadeem

MCQS

Question and Answers

Hibernation?

Animals can go into a state of reduced metabolism that may last several months. If this state occurs mainly in winter, it is called hibernation

Symbiosis example?

There are three different types of **symbiotic** relationships: mutualism, commensalism, and parasitism. Mutualism: both partners benefit. An **example** of mutualism is the relationship between the Egyptian plover and the crocodile. ... Commensalism: only one species benefits while the other is neither helped nor harmed. **(Internet)**

Example of foundation species?

Foundation species are sometimes referred to as 'Ecosystem Engineers'. **Examples of Foundation Species** are; Corals, Earthworms, Beavers and many more!

Nutrition and nutrition cycle?

A **nutrient cycle** refers to the movement and exchange of organic and inorganic matter back into the production of living matter. The process is regulated by the food web pathways previously presented, which decompose organic matter into inorganic **nutrients**. **Nutrient cycles** occur within ecosystems.

Factor of environment?

Environmental factor. An **environmental factor**, ecological **factor** or eco **factor** is any **factor**, abiotic or biotic, that influences living organisms. Abiotic **factors** include ambient temperature, amount of sunlight, and pH of the water soil in which an organism lives.

Reproduction of amoeba?

Asexual reproduction - • Binary fission (ppt)

Amoeba reproduces by the common asexual reproduction method called binary fission. After replicating its genetic material through mitotic division, the cell divides into two equal sized daughter cells. (Internet)

Mistake in natural selection?

A major driving force of evolution comes from **mistakes** made by cells and how organisms cope with the consequences, UA biologists have found.....Charles Darwin based his groundbreaking theory of **natural selection** on the realization that genetic variation among organisms is the key to evolution.

Characteristic of protozoa?

• No cell wall • Free-living or parasitic • Consumers • Habitat – aquatic (ponds,lakes, oceans) • Examples: amoeba, paramecium, dinoflagellates

Landscape element?

The ecosystems making up a landscape generally form a mosaic of visually distinctive patches. These patches are called landscape elements.

Nutrient influence by human and terrestrial ecosystem?

Human activities are significantly and rapidly altering the form and function of **terrestrial ecosystems**. For example, we are changing the chemical composition of the atmosphere, converting natural landscapes to urban areas, and transporting floral and faunal species far beyond their natural boundaries. **(Internet)**

What is hibernation and example?

Hibernation is a state or condition of inactivity of a plant or animal spending the winter in a long snooze especially for the purpose of rest. **Examples:** Reptiles and amphibians, bees etc.

Population growth?

Population growth is the **increase** in the number of individuals in a **population**. Global human **population growth** amounts to around 83 million annually, or 1.1% per year.

What is genetic drift?

Many changes in genotype are neutral or benign to phenotype and survival; these "cryptic" changes over time may result in genetic drift,

Landscape ecology?

"Landscape ecology is concerned with spatial patterns in the landscape and how they develop, with an emphasis on the role of disturbance, including human impacts" (Smith and Smith). It is a relatively new branch of ecology, that employs Global Information Systems. The goal is to predict the responses of different organisms to changes in landscape, to ultimately facilitate ecosystem management

Nutrient enrichment human?

Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems.
Nitrate concentration and export by the earth's major rivers correlate directly with human population density.
Human disturbance also increases export of ***phosphorus*** from aquatic catchments.
Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems

Protein synthesis of Organelles?

That is, some of the organelles • within eukaryotic cells are capable of protein synthesis. • These organelles • with their own genetic material • and protein-synthesizing capabilities • are thought to have been free-living bacteria • that entered into a symbiotic relationship, • eventually giving rise to eukaryotic cells

Optimal Foraging theory?

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

Factor affecting decomposition of environment?

The rate of **decomposition** is governed by three sets of **factors**—the physical **environment** (temperature, moisture and soil properties), the quantity and quality of the dead material available to decomposers, and the nature of the microbial community itself.

Troper?

A state of low metabolic rate and lowered body temperature, when food is scarce and night temperatures cold. Other animals can go into a state of reduced metabolism that may last several months.

Landscape Elements

A landscape is a heterogeneous area composed of several ecosystems.

•The ecosystems making up a landscape generally form a mosaic of visually distinctive patches.

•These patches are called landscape elements.

•Landscape ecology is the study of landscape structure and processes.

Work of Clausen, Keck, and Hiesey

Clausen, Keck, and Hiesey explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.

Long 5 Marks

Nutrient loss from ecosystem?

Major pathways in which these **nutrients** are **lost** include: soil erosion, leaching and gaseous **losses**. Leaching is a physical process where **nutrients** exit terrestrial **ecosystems** in the downward flow of water through the soil. ... Another **loss** of **nutrients** from **ecosystems** is due to denitrification.

Lamark contribution?

Lamarck is best known for his **contributions** to evolution, or Lamarckism, which suggests organisms acquire or lose traits based on how much they use them in their lives.

Key stone speices?

A **keystone species** is often a dominant predator whose removal allows a prey population to explode and often decreases overall diversity. Other kinds of **keystone species** are those, such as coral or beavers, that significantly alter the habitat around them and thus affect large numbers of other organisms.

Importance of bacteria?

The ecosystem, both on land and in the water, depends heavily upon the activity of **bacteria**. The cycling of nutrients such as carbon, nitrogen, and sulfur is completed by their ceaseless labor.

Food Web?

A food web is the natural interconnection of food chains and a graphical representation of what-eats-what in an ecological community. Another name for food web is consumer-resource system.

Role of bacteria in environment?

The cycling of nutrients such as carbon, nitrogen, and sulfur is completed by their ceaseless labor.....Decomposition is the breakdown of these organisms, and the release of nutrients back into the **environment**, and is one of the most important **roles** of the **bacteria**.

Foundation species?

Foundation species is used to refer to a **species** that has a strong role in structuring a community. A **foundation species** can occupy any trophic level in a food web (i.e., they can be primary producers, herbivores or predators).

Importance of protists?

The slime molds are important to ecosystems because they are decomposers, which release vital nutrients back into the environment. You could say life on Earth depends on the protists because they supply us with **oxygen**, are critical organisms in food chains, and recycle important nutrients for other life forms to use.

Sigmoidal population growth / movement

•Sigmoidal population growth can be modeled by the logistic growth equation, a modification of the exponential growth equation that includes a term for environmental resistance. In the logistic model, the rate of population growth decreases as population density increases. Research on laboratory populations indicates that zero population growth at carrying capacity may be attained by many combinations of reduced birthrates and increased death rates.

How Birds and mammals / Animals regulate body temperature

Birds and mammals rely heavily on metabolic energy to regulate body temperature. The physical nature of the aquatic environment reduces the possibilities for temperature regulation by aquatic organisms.

Most endothermic aquatic species are air breathers. Some organisms, mainly flying insects and some large marine fish, improve performance by selectively heating parts of their anatomy. The energetic requirements of thermoregulation may influence the geographic distribution of species.

Mode of Feeding of Amoeba

Feeding structures- •food vacuole •digestive enzymes •diffusion **(ppt)**

Amoeba feeds on microscopic organisms such as single-celled algae and bacteria. When the amoeba encounters a suitable organism, the cytoplasm flows round the prey and engulfs it, with a drop of water, in a food vacuole. The cytoplasm secretes enzymes into the food vacuole. **(Internet)**

Logistic population growth

•As resources are depleted, population growth rate slows and eventually stops; this is known as logistic population growth.

Long 10 Marks

Define Codon, nonsense codon and describe the relationship between genotype and phenotype? Codon

A **codon** is a sequence of three DNA or RNA nucleotides that corresponds with a specific amino acid or stop signal during protein synthesis.

Nonsense codon

Nonsense codon. A **codon** which does not code for any amino acid, but signals a termination of translation, or punctuation. The three **nonsense codons** are UAG (amber), UAA (ochre), and UAG (opal).

Nonsense mutation. A mutation which replaces a **codon** for an amino acid with a **codon** for chain termination (UAG, UAA, or UGA).

Relationship between genotype and phenotype

The **genotype** of an organism is defined as the sum of all its genes. The **phenotype** of an organism is the observable physical or biochemical characteristics of an organism, determined by both genetic make-up and environmental influences.

Gene and its characterization?

• The idea of the gene has evolved through the history of genetics • We have considered a gene as • A discrete unit of inheritance • A region of specific nucleotide sequence in a chromosome • A DNA sequence that codes for a specific polypeptide chain • In summary, a gene can be defined as a region of DNA that can be expressed to produce a final functional product, either a polypeptide or an RNA molecule **(PPTS)**

Characterization refers to **the** detection of variation as a result of differences in either DNA sequences or specific **genes** or modifying factors. **(Internet)**